NUCLEAR RESEARCH AND TRAINING ACTIVITIES FOR THE MFF 2028-2034 (FP10)

AN STC OPINION

Euratom Scientific and Technical Committee

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FOREWORD

The Euratom Scientific and Technical Committee (STC) is, together with the European Economic and Social Committee and the Committee of the Regions, one of the advisory bodies established by the European Treaties. Since 1958, it has delivered opinions on relevant scientific and technical issues, in particular, in relation to the Euratom research and training programme (henceforth 'the Euratom programme').

The members of the STC are high-level nuclear and radioprotection experts, appointed in a personal capacity by the Council of the European Union. They serve a five-year term with the current mandate of the Committee ending in November 2028.

This document was adopted by the STC with Carlos Varandas as Chair and William D'haeseleer as Vice-Chair and Rapporteur on this document.

The preparation of this document was facilitated by the STC Secretary, Frederick Mariën, and his team, Franklin Mamo and Tomasz Śliwiński.

ACRONYMS AND SPECIFIC TERMINOLOGY USED

AI	Artificial Intelligence
BNCT	Boron Neutron Capture Therapy
DTT	Divertor Tokamak Test facility
ENSREG	European Nuclear Safety Regulators Group
ERC	European Research Council
EU	European Union
EU-FP10	The EU's prospective 10th Framework Programme for research and innovation (2028-2034)
GIF	Generation IV International Forum
HERCA	Heads of the European Radiological Protection Competent Authorities
IAEA	International Atomic Energy Agency
IEA	International Energy Agency
ITER	International Thermonuclear Experimental Reactor
JRC	Joint Research Centre
LCA	Life Cycle Assessment
LTO	Long Term Operation
NEA	Nuclear Energy Agency of the OECD
OECD	Organisation for Economic Co-operation and Development
PPP	Public-Private Partnership
R&D	Research and Development
RD&I	Research, Development and Innovation
RPT	Radio-Pharmaceutical Therapy
SMR	Small Modular Reactor
STC	Scientific and Technical Committee
TRL	Technology Readiness Level
WENRA	Western European Nuclear Regulators Association

AIM OF THE DOCUMENT AND THE STC'S ROLE

This Opinion presents key recommendations from the STC as required by Article 7 of the Euratom Treaty in the Commission's preparation of a proposal on nuclear research and training during the Multiannual Financial Framework (MFF) for the period 2028-2034.

As outlined in the Treaty, the STC members "shall not be bound by any other instructions". The STC does not advocate for nuclear energy as such, but it does provide unbiased, genuine science-based advice on how nuclear science and technology[1] are related to the broader issues like the economy, the protection of human health & the environment and the medical & industrial applications of ionising radiation.

While the EU's Member States may hold differing views on nuclear energy, it's in the EU's and all Member States' interest to have access to informed advice. Nuclear technology deployed in Member States or outside the EU can have cross-border impacts.

Nuclear is now seen as playing a more important role in the green transition while the opportunities it offers in other areas such as medicine are increasingly being appreciated. Comprehensive nuclear research is crucial for the EU to stay at the forefront of scientific and technological advancements. This includes improving and optimising current practices and launching new initiatives on nuclear systems, fuel cycles, safety, waste treatment and management, radiological understanding and medical and industrial applications of ionising radiation.

The broader political and policy context, such as the Commission's preparation for a fusion strategy in 2025, is still fluid. A future follow-up to this opinion might therefore be necessary.

THE GLOBAL CONTEXT IN WHICH THE EU RESIDES

The STC observes and recognises that the events of the last few years have made the world more fragmented and conflicted. This will require a fundamentally different approach in the future to that taken in the last three decades so as to ensure the EU's welfare and security.

The EU should focus on innovation to strengthen its economy, improve living standards and ensure social and health progress. Europe's reinvention must be driven by research, development and education. Ensuring security of energy supply is increasingly complex and continued adherence to global decarbonisation agreements, while of paramount

[1] Throughout this text, the wording 'technology' is used in its original sense (broadly related to applied sciences and engineering); it is not limited to the 'information, communication, data and/or artificial intelligence' sectors.

importance, remains uncertain. Achieving these goals is proving more expensive than anticipated, though there's hope that it may become more affordable over time and almost certainly when factoring in the avoided costs of climate change impacts. The COVID-19 pandemic and its impact on supply chains highlighted the risks of overreliance on external trade and our lack of resilience to unexpected events. Moreover, geopolitical tensions and protectionist tendencies continue to strain the global energy landscape and honest international intellectual exchanges.

The STC notes that its perspective aligns with the key themes of the Draghi report[2] which proposes three key areas for action to reignite growth:

- Closing the innovation gap with the US and China via collective efforts, especially in advanced technologies.
- Developing a joint plan for decarbonisation and competitiveness.
- Strengthening security and reducing dependencies.

But the Draghi report highlights three key barriers standing in our way:

- Lack of focus: Europe sets common goals but fails to establish clear priorities or coordinated actions.
- Wasting resources: Despite significant collective spending, resources are spread thin with insufficient collaboration on innovation.
- Poor coordination: Europe lacks a coordinated industrial strategy that combines fiscal, trade and foreign economic policies to secure supply chains.

The Draghi report also notes that new advanced nuclear technologies could play a crucial role in integrated energy systems with high penetration of renewable energy sources and could contribute to building a competitive technological supply chain in the EU.[3] While the true test of this statement lies in the outcomes, it aligns with the International Energy Agency's (IEA) view that, without ignoring the challenges it presents, nuclear power is likely to be part of decarbonisation efforts in many countries worldwide, thanks to the opportunities it offers.[4]

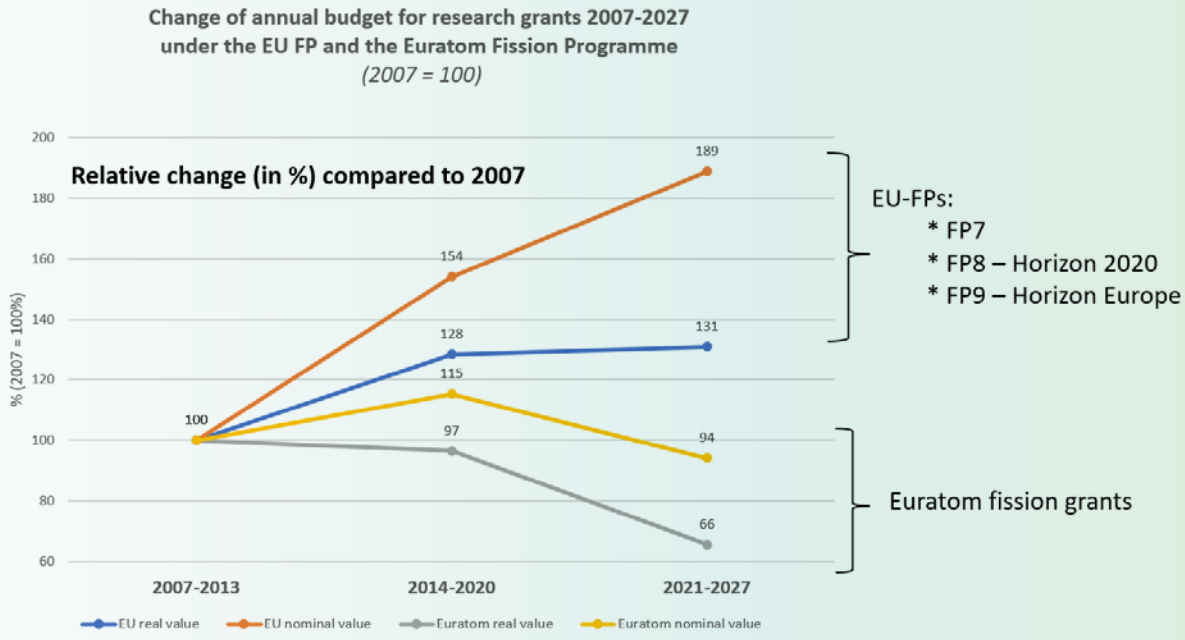
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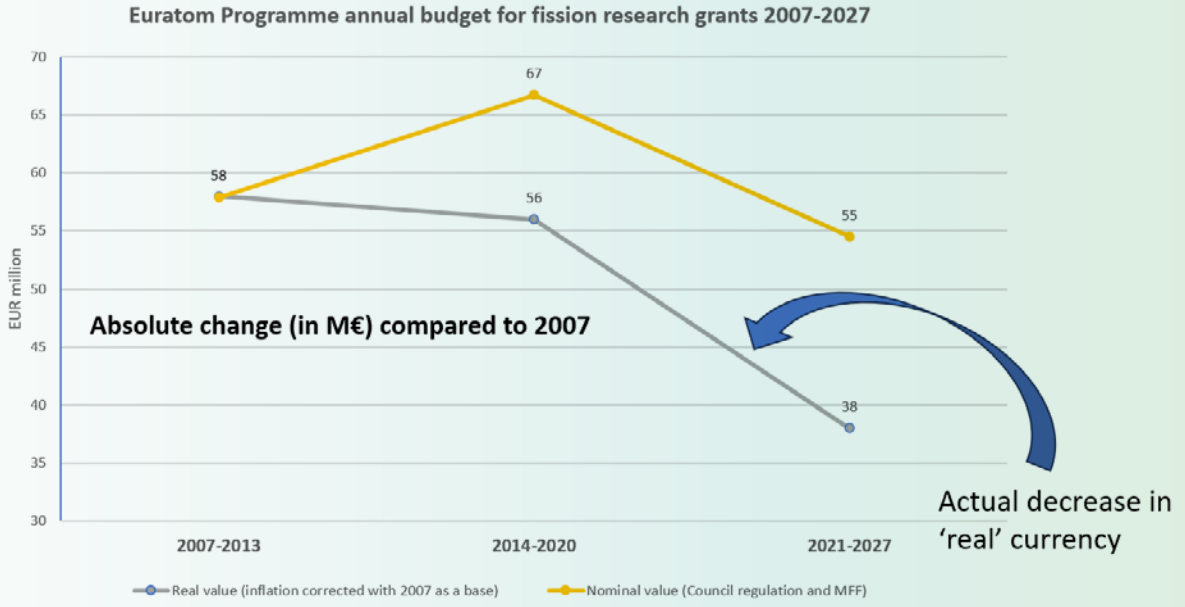
[2] *The future of European competitiveness* (September 2024) https://commission.europa.eu/topics/strengthening-european-competitiveness/eu-competitiveness-looking-ahead_en.

[3] Draghi Report, 2024, Part B, p 20, Box4.

[4] See e.g., the sequence of IEA reports https://www.iea.org/reports/nuclear-power-in-a-clean-energy-system; https://www.iea.org/reports/nuclear-power-and-secure-energy-transitions; https://www.iea.org/reports/the-path-to-a-new-era-for-nuclear-energy.

It is regrettable that the Euratom programme's funding for fission energy research has significantly decreased over the last decades, undermining expertise. The figures below show that the decline is in both relative and absolute terms.[5]





Source: European Commission

Current resources do not align with the goal of fostering a new dynamic and fail to support the necessary research and education required to address the EU's energy transition and strategic autonomy challenges.

[5] The graphs do not include funding for the Joint Research Centre (for the so-called 'direct actions').

The STC is of the view that the EU has lost its leadership position in nuclear fission, which could negatively impact the ongoing operation and maintenance of existing nuclear facilities as well as the development of new facilities. Reestablishing this leadership is crucial for Europe's global influence and role in nuclear safety and in setting radiation protection standards.

The Draghi report calls for a substantial increase to the overall European research and innovation budget. Given the current and future challenges, it isn't unreasonable for the STC to firmly requests a 'correction' and recommend that the EU Budgetary Authority, without hesitation and swiftly, substantially increase the new Euratom programme's research, development and innovation (RD&I) budget. Lacking sufficient RD&I funding risks hampering the ambitions the EU has set for itself. There is a pressing need for adequate funding to constructively and critically examine the merits and the potential of novel carbon-free nuclear technologies expected to be deployed in several member states. The EU could face significant consequences for failing to optimise the energy trilemma: deliberate, effective and efficient climate action, a comfortable security of energy supply, and affordability for citizens and the competitiveness of our industry, energy-intensive companies in particular. The arguments for the nuclear medical field are equally compelling, and the needs are just as urgent.

In order to achieve the established objectives, it is essential to significantly reduce, dismantle, or, at the very least, simplify ineffective and counterproductive administrative barriers. This is particularly relevant to the energy transition, but it also extends to medical and industrial initiatives. While maintaining the integrity of well-considered environmental and safety regulations, the issue of 'permitting' demands careful and thorough attention. In the nuclear field—encompassing fission, fusion, as well as medical and industrial applications—greater efforts are required to achieve convergence towards common EU guidelines. Accordingly, the Euratom programme must pursue a result-oriented effort to align regulatory safety and radiation-protection standards within the EU. Further harmonisation in nuclear safety approaches is needed, aiming to develop binding uniform regulatory codes and appropriately strict but realistic standards at the EU level and beyond (building on the HARMONISE project).[6]

Given the EU's refocused industrial policy with its increased emphasis on valorisation and innovation, the Euratom programme's budget should continue supporting basic research while increasing efforts to bridge to applied research and to foster transdisciplinary industrial research initiatives.

NUCLEAR KNOWLEDGE, SKILLS AND COMPETENCES

Strategic autonomy requires enhanced European collaboration to boost education and training in the nuclear and radioprotection fields, ensuring a skilled workforce. Lack of investment in training, education, research, and infrastructure may especially compromise radiation risk management.

[6] https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/projects-details/43298916/101061643

- The imminent retirement of a significant number of nuclear experts requires urgent replenishment.
- Experts with deep nuclear knowledge are crucial in understanding and dealing with future nuclear facilities, including medical and industrial radiation equipment.
- Archived documents and information from operating, shutdown and dismantled nuclear facilities in Europe must first be evaluated by professional nuclear experts and after selection transferred to databases, potentially supported by artificial intelligence (AI) tools. The Euratom programme should actively engage in 'knowledge management and preservation' efforts within the EU together with Nuclear Energy Agency of the Organisation for Economic Co-operation and Development (OECD/NEA) and the International Atomic Energy Agency (IAEA).

The STC recommends that the Commission address education and training holistically, for the fission, fusion and non-energy sectors to create a European nuclear competence area, supported by a strategic plan with clear milestones. A strategic plan on education should identify areas needing significant funding for clear outcomes, alongside initiatives that can progress with more modest financial support.[7] The STC insists that attracting new talent to the nuclear field would be a major challenge for the EU's education and training strategy. The Euratom programme's support for nuclear education depends on offering clear career prospects, incentives and opportunities, which should be better promoted across research centres, industries and government agencies. Following the energy crisis, young people are more open to the ideas of nuclear fission and fusion and this has likely been helped by its inclusion in the taxonomy of sustainable investments.

The effort required to effectively harmonise future EU safety and radioprotection regulations—ensuring they are both well-balanced and appropriately strict—must include timely educational initiatives targeted at nuclear regulators. These initiatives should involve common training programmes and knowledge transfer related to emerging scientific and technological developments in new systems. Emphasis should be placed on collaborative projects between industry and regulatory authorities, including the development of shared training materials for both technical and legal experts. The STC urges the Commission to intensify dialogue with the European Nuclear Safety Regulators Group (ENSREG), the Western European Nuclear Regulators' Association (WENRA) and the Heads of the European Radiological Protection Competent Authorities (HERCA) to obtain feedback for shaping research towards producing useful knowledge for regulators.[8]

[7] The strategic plan could build on the work of the recently established Skills4Nuclear Project (2025-2028); see https://cordis.europa.eu/project/id/101213280.

[8] https://www.ensreg.eu/; https://www.wenra.eu/; https://www.herca.org/

A future without applications of AI is inconceivable. AI is expected to permeate all areas of science, technology, and society wherever electronic data processing & control systems are employed. It is therefore imperative that the Euratom programme actively supports education and training initiatives focused on the effective and responsible use of AI. These efforts must also address the potential risks associated with unquestioned and/or uninformed application. Such teaching and training activities should target designers of nuclear reactors (both fission and fusion), developers of nuclear-related medical and industrial technologies, current and future operators, as well as regulatory authorities.

The STC advocates for enhanced shared initiatives, whereby the Euratom programme complements national nuclear skill strategies, e.g. via promoting transnational access to nuclear research infrastructures like the 'OFFERR' project[9] and support for mobility and education through platforms like ENEN2Plus.[10]

Public-Private Partnerships (PPPs) are crucial to fostering a new generation of nuclear fission and fusion energy scientists and engineers. Internships with industry[11] and highlighting career opportunities can help attract top-tier talent to the sector.

Finally, nuclear researchers' eligibility should be extended to all Marie Skłodowska-Curie Actions not just Postdoctoral Fellowships.

RADIOACTIVE WASTE MANAGEMENT AND FINAL DISPOSAL

The STC supports research and development (R&D) efforts to minimise the waste requiring storage and/or final disposal. The approach to life-cycle assessment (LCA) should be comprehensive; it should start from the design phase of a facility and encompass resource extraction to facility decommissioning and waste management. This should include high-, medium- and low-level waste and recycling and apply to all nuclear-related facilities. Financial support for developing a common European nuclear LCA methodology is recommended.

The EURAD-2 Partnership[12] should evolve to support Member States in designing and implementing national waste management programmes as required by the Radioactive Waste Directive[13]. It should address the changing waste landscape, including mature

[9] https://snetp.eu/offerr/; https://enen.eu/index.php/portfolio/offerr/

[10] https://enen.eu/index.php/portfolio/enen2plus-project/

[11] Throughout the remainder of this document, 'industry' includes well-established nuclear companies up to private equity (and/or venture capital) funded start-ups.

[12] https://www.ejp-eurad.eu/

[13] Council Directive 2011/70/Euratom of 19 July 2011 establishing a Community framework for the responsible and safe management of spent fuel and radioactive waste (OJ L 199, 2.8.2011, p. 48, ELI: http://data.europa.eu/eli/dir/2011/70/oj).

repositories and non-nuclear Member States' needs. It should play a strategic role in research and in providing tools for radioactive waste repositories and addressing safety concerns. Efforts should also focus on engaging regulators in EURAD-2 in ways that suit different national frameworks. Knowledge management is crucial due to limited staff and turnover challenges.

Maintaining civil society involvement in the EURAD-2 R&D Partnership is crucial for informed decision-making, whether through applying and enhancing existing techniques (such as the so-called 'double wing' approach[14] and a 'public information and participation' —PIP— exercise) or by developing new ones.

Joint fusion-fission studies on waste issues are crucial for sharing methodological developments and costs. The fusion programme should focus on issues like tritium contamination, maintenance of activated structures, radioactive dust from reactor materials and mobilisation of corrosion products. A comprehensive waste strategy must be included from the design phase in fusion facilities.

RADIATION APPLICATIONS AND PROTECTION

Research is important in advancing therapeutic medical applications, optimising current treatments and supporting the adoption of new techniques. This research is crucial for human health and benefits all of EU society:

- As cancer survival rates improve, patients may develop secondary cancers or second primary cancers, requiring multiple radiation treatments. The case for reirradiation must balance treatment efficacy with the risk of toxicity. Further research is needed on radiation tolerance, optimal reirradiation doses and managing past treatment doses.
- Radiopharmaceutical therapy (RPT) development is active but knowledge of its impact on human tissue is limited. Research on the radiobiology of internal exposures from RPT should be intensified.[15]
- FLASH therapy delivers ultra-high dose rate radiation, with the potential to reduce toxicity in healthy tissues without compromising anti-cancer effects. Further research on its radiobiology is crucial for safe patient implementation and preventing adverse events.
- Boron neutron capture therapy (BNCT) is not widely available in Europe due to limited access to neutron facilities although accelerator-based facilities are emerging as a clinical option. Interest in BNCT in the Euratom programme would be welcome.

[14] 'Double Wing' approach: Integrating both 'technical' and 'societal' considerations to ensure that the technical aspects of radwaste management are aligned with the social acceptance and trust of the public, fostering a more holistic and sustainable solution.

[15] In line with the recently published roadmap of the Eurados report 2024-03.

While AI has a wide range of applications, it holds significant potential to accelerate scientific radioprotection research and the EU should lead its development and application. AI can advance personalised medicine and the STC advocates for an EU-wide strategy to optimise radiation protection in healthcare. This strategy should be cross-disciplinary, co-funded and address recommendations from the Euramed RoccNRoll agenda,[16] with a focus on educating medical professionals in AI.

Many questions remain about the effects of low and high radiation doses on human tissue. The Euratom programme should support large-scale databases and dose registries in Europe to include medical examinations (nuclear medicine diagnostics and x-ray imaging) and radiation therapies (e.g. radiopharmaceutical therapy, particle therapy, FLASH therapy and BNCT). These registries should combine AI with traditional systems for high-quality clinical and radiobiological research.

As with waste management issues, radiation protection should be integrated from the start in the development of new energy technologies, including modular reactors. This can be achieved through co-funding opportunities or funding requirements in projects and partnerships.

The PIANOFORTE partnership[17] has made significant progress and should continue to be supported. However, further enhancement of its structure is needed to ensure its research remains relevant and addressed to the key challenges. Radiobiology, AI, databases and registries could be included in the portfolio of this enhanced PIANOFORTE. The Commission should continue launching projects on major issues, taking care to avoid duplication.

As recommended by the experts contributing to the interim evaluation of the 2021-2025 Euratom programme, a systematic analysis of national radiation protection programmes should be made, the outcome of which could serve as valuable input in preparing future Euratom programmes and calls.

The STC suggests improving stakeholder networking with longer-term perspectives through PIANOFORTE, simplifying the legal and financial processes for funding and involving more third parties like universities, hospitals and medical facilities.

Dedicated funding mechanisms with other relevant partnerships and sources, such as with the next EU Framework Programme (EU-FP10) and the European Space Agency, should be established to support transdisciplinary research and co-funding for dynamic projects.

[16] https://roccnroll.euramed.eu/

[17] https://pianoforte-partnership.eu/. PIANOFORTE refers to the "Partnership for the Integration of Advanced Radiation Protection Research in Europe"; it encompasses a.o. the six European research platforms in radiation protection (MELODI, EURADOS, EURAMED, NERIS, ALLIANCE and SHARE). The word 'pianoforte' is not an abbreviation or acronym.

NUCLEAR FISSION ENERGY

Waste management, the fuel cycle (front- and back-end), new nuclear systems, infrastructure, simulation and the long-term operation (LTO) of existing reactors are important areas for research.

For the benefit of all EU Member States, it is essential that the post-2027 Euratom programme continues to uphold the highest safety standards and judicious waste management principles.

However, with the wide array of innovative and cutting-edge concepts emerging[18] the programme should go beyond supporting projects focused solely on safety and waste management in the narrow sense. In particular, adequate efforts are urgently needed now to understand the underlying reactor-system physics & technological principles and the physico-chemical fundamentals for robust waste management.[19] Therefore, active but supplementary support from the Euratom programme to the development of next-generation reactors is needed to truly understand and stay ahead of novel science and engineering solutions. The Euratom programme can play a supportive role in enabling the emergence of advanced reactors in the EU that deliver on their promised performance while adhering to the highest safety requirements. Encouraging and stimulating innovation and contributing to technological breakthroughs from European industry will be essential to secure a diverse, resilient and sustainable energy mix for the EU.

The EU could certainly benefit from the Euratom programme's co-funding collaborative research with the European Industrial Alliance on Small Modular Reactors (SMRs) to keep the focus on EU-relevant and deployable concepts.[20] Crucially, the Euratom programme cannot act alone; its role is supplementary to industry, which must take the lead, and Member States, which must provide the necessary policy support. Co-funded Euratom projects could remove some technological bottlenecks while involving more private financing and industry experience. Taking a more extensive but vigilant approach to industrial developments would facilitate the exploration and clarification of the presumed role that new, smaller and advanced nuclear reactors could play in ensuring a secure supply of electricity and heat, complementing renewables through seasonal fluctuations or when flexibility and geographical exchange of instantaneous electrical power are limited or obstructed. The STC therefore recommends launching joint indepth studies on the co-existence in national electricity grids of renewables having a large but variable share of the energy mix and nuclear. These studies should explore the complementarity and competition between these generation means, including their impact on the overall European electric power system and on countries heavily reliant on electricity import and export.

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[18] See e.g. the NEA/OECD "Small Modular Reactor Dashboard" (2nd Ed), at https://www.oecd-nea.org/jcms/pl_90816/the-nea-small-modular-reactor-dashboard-second-edition?details=true or the IAEA "Advances in Small Modular Reactor - Technology Developments", at https://aris.iaea.org/Publications/SMR_Booklet_2022.pdf

[19] This may include aspects such as fast-neutron-spectrum reactor physics (reactor kinetics and dynamic control & stability), thermal hydraulics of non-standard coolants, materials behaviour, fuel cycle characteristics upstream & downstream. This would encompass theoretical modelling and experimental analysis for validating the models.

[20] European Industrial Alliance on SMRs: https://single-market-economy.ec.europa.eu/industry/industrial-alliances/european-industrial-alliance-small-modular-reactors_en

The comprehensive and meticulous studies on the EU-wide electricity system (preferably in collaboration with the EU-FP10) should assess how the envisaged complementarity of nuclear and renewables impacts the competitiveness of the EU's entire economy, especially in energy-intensive sectors like process industries and data centres.[21]

As to reactors, R&D support from the Euratom programme should focus on the more advanced innovative concepts. Further effort is needed to address the main safety and regulation issues at an early stage of their design. To be more effective and efficient, Euratom projects in this area should concern only a few technologies with the strongest industrial interest and coordination.

An important factor in ensuring safe deployment of new reactors is the licencing process. The Euratom programme should support closer cooperation between national regulators to help establish a shared and coherent approach among regulators to safety requirements for innovative reactors.

The STC argues that the EU cannot afford to stay passive in the face of the pursuit for more advanced nuclear technologies as it risks having to import non-EU nuclear designs with very limited influence over safety standards, quality control and reliable supply chains. This is also in the interest of EU Member States that have avoided domestic nuclear generation, as they benefit from cross-border power exchanges of nuclear origin whilst possibly affected in case of accidental radioactivity releases. Reactor and fuel cycle studies should be prioritised equally, with some concepts aiming for fuel cycle closure to improve waste management and support circular economy goals, such as multi-recycling in Generation IV advanced systems.[22] More research is needed on advanced nuclear fuel treatment, including partitioning and transmutation, with a thorough evaluation of the claimed benefits. Additionally, efforts should intensify on developing new and innovative radioisotopes for industrial and medical use. The behaviour of HALEU fuel (5-20% enrichment), essential for research and advanced reactors, also requires deeper exploration, keeping in mind proliferation aspects. Euratom-funded research is furthermore required to develop accident-tolerant fuels.

New reactor systems, particularly modular ones, are expected to go beyond electricity generation by including (high-temperature) heat and hydrogen production. R&D on these technologies should be collaborative to speed up innovation, while also maintaining a critical perspective on fulfilling the promises made. The STC sees significant potential for Euratom- funded research providing solutions and knowledge for development and deployment of safe future reactor systems. The Euratom programme should support long-term research and industrialisation efforts, focussing on technologies with enhanced safety features. A key element is the effect/impact of integrating these new systems into the wider energy landscape. The STC suggests developing a joint European public-private strategy and creating use cases for different industrial sectors. R&D should leverage synergies between the Euratom programme and EU-FP10.

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[21] The hoped-for 'combined' Euratom & EU FP10 funded studies should especially be addressed to the nuclear industrial SMR alliance, large electricity consuming industry, and the electricity transmission system operators through e.g., ENTSO-E.

[22] The term 'Generation IV' has evolved and now encompasses more than the six concepts selected by the GIF (Generation IV International Forum — https://www.gen-4.org/). In summary, 'generic' Gen IV advanced systems are grounded in updated regulations and guidelines, with incorporation of complete passive safety (see e.g., https://www.amacad.org/sites/default/files/academy/pdfs/nuclearReactors.pdf) and more resource efficiency.

Essential nuclear research infrastructures must be identified and current and future shortfalls addressed, as Europe's nuclear research facilities are ageing. This could be based on the OFFERR project's facilities which emphasise the need for transnational access to nuclear research. A comprehensive strategy should be developed to indicate the required infrastructures for the future. The analysis should cover both fundamental research infrastructures for low-technology readiness level (TRL- science and technological infrastructures at intermediate TRLs[23], bridging research and practical applications. A strategic approach is needed to anticipate technological infrastructure needs and foster industry collaboration for the market adoption of key technologies.

To enhance Europe's strategic autonomy in simulation codes and high-quality data, the STC recommends giving particular attention to the digital aspect, encompassing its various components: code development, data acquisition & control, processing, evaluation, verification & validation, storage and utilisation. Efforts must be intensified in the field of modelling to better reflect the fundamental phenomena, emphasising the development of coupled multi-model, multi-scale and multi-physics approaches (such as neutronics, thermal-hydraulics, and mechanics), along with the creation of codes tailored to new supercomputing architectures and AI methods. A key objective is to minimise modelling uncertainties. These efforts should be closely coordinated with experimental research, as simulation quality relies on reliable experimental data. Clearly, the Joint Research Centre (JRC) and its nuclear research activities should be supported in these efforts.

As Member States extend the operational lifetimes of nuclear reactors, there is need for research on LTO, focusing on plant safety, material ageing, modelling and nuclear data, that the Euratom programme should support. Collaborative projects with industry, regulators and their Technical Support Organisations (TSOs) are essential, along with access to experimental infrastructures. However, LTO investments in the to-be refurbished nuclear facilities should be financed by their owners. R&D on severe accidents at existing plants should be strengthened, with greater European coordination, to identify weaknesses in refurbished units and reduce uncertainties.

To foster breakthrough, high-risk or innovative ideas, the Euratom programme should introduce new calls for projects, such as bottom-up or blue-sky initiatives. One approach to achieving this is to include applied nuclear science and technology topics within the scope of the European Research Council (ERC), in particular if ERC funding is increased. Additionally, cross-disciplinary initiatives between nuclear and non-nuclear research, with budgets shared across different EU programmes, should be encouraged. Examples include collaborative actions on materials, nuclear data, thermal-hydraulics and physical-chemical processes. This applies to fission-related subjects, the plasma physics & fusion area, the nuclear medical field and even inventive/ingenious applications in a broader industrial context. These synergies should be aligned with EU-FP10 instruments.

[23] TRL stands for Technology Readiness Level. See, e.g., https://ec.europa.eu/research/participants/data/ref/h2020/other/wp/2018-2020/annexes/h2020-wp1820-annex-g-trl_en.pdf. A further step should be what can be referred to as Market, Commercial or Business Readiness Level, abbreviated as MRL, CRL or BRL, respectively.

NUCLEAR FUSION ENERGY

Recent advances in nuclear fusion are promising but highlight the need for a revised R&D programme, focusing on commercialisation. Whilst still pursuing better scientific & technical understanding and concurrently striving for optimisation & refinement, fusion R&D must shift from a science-only exploration to a more applied energy-driven endeavour, with more industry involvement. Addressing plasma science challenges and developing safe technological solutions for reliable fusion devices is crucial. Close collaboration between public research institutions and private industry in the EU is needed, along with a reassessment of the EU fusion programme's organisation and management to accelerate progress.

The EU fusion programme is at a crucial juncture, with ITER updating its timeline amid growing international competition from countries like the USA, UK, China, Korea and Japan. Europe should adopt a 'parallel' strategy for exploring relevant and legitimate alternative pathways to accelerate technology validation, despite the higher risks to success. This risk is however a necessary price for fusion acceleration. Negative outcomes can drive innovation and lead to better solutions.

Despite delays, ITER remains a critical milestone especially in understanding tokamak-related instabilities and particle exhaust. However, during ITER's construction and operation, complementary parallel strategies should also be pursued:

- Intensive experimental campaigns to improve plasma performance in both tokamak and stellarator devices must be pursued in existing European facilities.
- Technological component development for future fusion reactors should progress by raising lower TRLs to higher, validated and trustworthy TRLs.
- A practical strategy for rapid validation involving a 'modular' approach. Instead of testing a large, complex component, individual parts can be tested under harsh reactor-like conditions in small-scale testing setups. Examples include the IFMIF-DONES[24] material testing facility, the potential Volumetric Neutron Source and the Divertor Tokamak Test (DTT) facility.

It is important to reflect on a two-phase approach for the next-step fusion experiment or pilot plant. The first phase would use a partially completed facility to test large components, followed by finalising it for electricity generation demonstration.

Accelerating reactor studies for realistic pilot fusion plants is essential, focusing on the full integration of reactor components and safety-related licensing. While tokamak-based studies should lead, significant attention should also be given to the stellarator approach. Regular mutual design reviews between tokamak and stellarator experts, with the involvement of plant-design industries and external experts, are crucial.

[24] IFMIF-DONES stands for International Fusion Materials- Irradiation Facility – DEMO Oriented Neutron Source, https://ifmif-dones.es

The EU fusion programme should maintain its focus on magnetic confinement. An appropriate watching brief on the area of inertial fusion is nevertheless desirable, as well as accelerating the development of key mutual technologies in magnetic and inertial fusion Furthermore, it is essential to monitor progress in new, credible innovative approaches.

To support the parallel approach, a strategic research and innovation agenda should be developed, identifying key technological components and addressing scientific gaps. This agenda must align with the European Fusion Strategy and Roadmap.

The EU must master key fusion technologies and develop associated industries and supply chains. Vulnerable areas of dependency should be examined and, where there are gaps, co-development with industry and labs should be considered. Strategic partnerships should be formed to avoid monopolistic situations.

The construction of ITER has underscored the critical importance of quality control as well as strict adherence to the safety standards mandated by licensing authorities. An in-depth study should be initiated to facilitate discussions with regulators on nuclear fusion standards particularly regarding the entire tritium fuel cycle, neutron-activated and tritium-contaminated reactor components and shielding. This study should aim to identify the differences with fission, while ensuring that overall safety, security and waste management requirements are not compromised. It is imperative to establish uniform EU-wide standards, which are consistent, yet appropriately stringent in line with the radiological risks. Ideally, this should pave the way for the adoption of global standards. The STC recommends enhancing industrial strategy and establishing EU frameworks to boost industrial participation in fusion. A market analysis of fusion technologies will support PPPs. Small and Medium-sized Enterprises (SMEs) should be more involved through Fusion for Energy (F4E)[25], bringing specialised expertise and innovation. Additionally, fostering technology and knowledge transfer between the fusion industry and other sectors, like space and data-driven research, is crucial for collaboration and progress.

International collaboration should not work against the EU's strategic autonomy but support it. The EU should engage in global fusion collaboration, particularly through the ITER project, while protecting intellectual property and maintaining geopolitical autonomy. In particular, participation in high-power and tritium-capable magnetic fusion experiments outside the EU should be encouraged to test reactor-relevant plasma scenarios in preparation of ITER's operation and the design of a European next-step device.

Choosing an EU site for the next-step European pilot plant would strengthen Europe's fusion energy programme, with strategic, technical, economic and political factors needing careful evaluation for long-term success and competitiveness.

[25] F4E is the joint undertaking managing Euratom's contribution to ITER.

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